

Spreading 39 Points in the Plane

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Abstract

We place 39 points in the plane so that the squared ratio of the largest to the smallest distance between them is at most 36.28378+. Erich Friedman’s table of best known configurations lists n up to 31. Deleting a point from our 40-point configuration gives 37.08127; this one is smaller. The ratio is computed exactly from the published coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 39 points in the plane with*

$$\left(\frac{\max d}{\min d}\right)^2 = 36.28378\dots$$

The exact value, a rational number, is given in Appendix A.

2 Context

Erich Friedman’s page *Minimizing the Ratio of Maximum to Minimum Distance* lists the best known configurations for $n \leq 31$. The value for $n = 39$ is not on the page. Deleting a point from our 40-point configuration gives 37.08127; this one is smaller.

3 Verification

The coordinates are decimals with 12 digits after the point, so they are exact rationals; all squared distances, and their largest-to-smallest ratio, are computed with integers. Run

```
python friedman-maxmin2d/verify.py friedman-maxmin2d/coords/n39.txt
```

in <https://github.com/Ge-limin/math>.

4 How it was found

Random starts of a constrained local optimizer (SLSQP), then basin hopping seeded from the neighbouring sizes (`friedman-maxmin2d/`).

A Exact value

```
numerator:
36283783730718313191580642
denominator:
999999999998904154194125
```

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

B Coordinates

39 points, x and y per line. File: friedman-maxmin2d/coords/n39.txt.

4.023859123326	-1.129385383375	6.806506184966	-2.065304977728
4.892333390553	-2.683223815236	2.990544644362	-0.434503473308
6.769914950857	-3.064635294283	4.135243230194	-3.336534223150
1.624645063106	-1.857748359370	6.350284997765	-3.972330563891
3.208253263604	-4.114330942161	1.675390597049	-0.173758980668
3.085351862378	-1.474644881649	2.188662148619	-1.031985241248
1.277244072339	-3.766641040921	6.146586419839	-0.220757676573
4.183518608016	-4.335368287395	2.477273092792	0.423722787272
4.430489587853	0.741030457941	4.766277954195	-5.148013071074
1.111373511536	-0.999522098790	5.656526762020	-4.692538527146
3.175069016904	-3.057132218624	3.943879619781	-0.132588875048
6.666119045635	-1.075208289970	5.808410432192	-2.126992181574
4.887408090366	-0.470848461642	3.771127254186	-5.246375077318
2.795861909774	-5.025337732084	2.521334776865	-2.300407999770
5.666376795257	-1.097911435306	4.856044792718	-1.683882463311
0.792945712410	-2.412974610597	5.376667822105	0.417384433669
3.933585073184	-2.353426198162	1.689635426169	-2.855634250997
5.072156803356	-3.876536105065	1.930978312378	-4.523365259775
3.430608068211	0.725637385532	5.774012811936	-3.155072729520
2.343369666208	-3.612358469851		